Resolving Wigner Negativity and Semiclassical Caustics via Finite Classical Action

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Wigner negativity and semiclassical caustics both follow from assuming unbounded classical action. Enforcing the finite action of a physical system via symplectic phase-space cells resolves both simultaneously. The resulting phase-space densities are provably non-negative and free of turning-point divergences.

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INTRODUCTION

Phase-space representations of quantum mechanics bridge classical trajectories and quantum amplitudes, but the mapping introduces two persistent pathologies. The Wigner–Weyl distribution exhibits negative values wherever quantum interference is strong [1], while the WKB probability density diverges at classical turning points where the local momentum vanishes [2, 3]. These are typically treated as separate inconveniences of separate formalisms.

They are not. Both pathologies are kinematic consequences of the same assumption: that classical action is unbounded. Wigner negativity arises from interference on scales below $\hbar/\Delta p$; the WKB caustic arises from demanding $\delta q \to 0$ at the turning point, which forces $\Delta p \to \infty$ and hence infinite classical action. No physical system satisfies this condition.

The standard remedy applies a fixed Gaussian kernel to yield the Husimi Q-function [4], which suppresses negativity and softens the caustics. But the Husimi kernel is set once by the ground state and carries no information about the system's classical action — a kernel decoupled from A cannot resolve structure governed by A, and introduces artifactual tails broader than true quantum tunneling.

Here we enforce what the physics already demands: a finite classical action A constrains the phase-space resolution to Zurek's reciprocal scales $\delta q = \hbar/\Delta p$ and $\delta p = \hbar/\Delta q$ [5]. Drawing on de Gosson's symplectic geometry [6, 7] and Gromov's non-squeezing theorem [8], we construct squeezed Gaussian kernels governed entirely by A. These kernels saturate the Heisenberg limit, provably eliminate Wigner negativity by Hudson's theorem [9], and bound the momentum support at turning points, replacing WKB divergences with finite-amplitude densities. Finite classical action resolves Wigner negativity and semiclassical caustics.

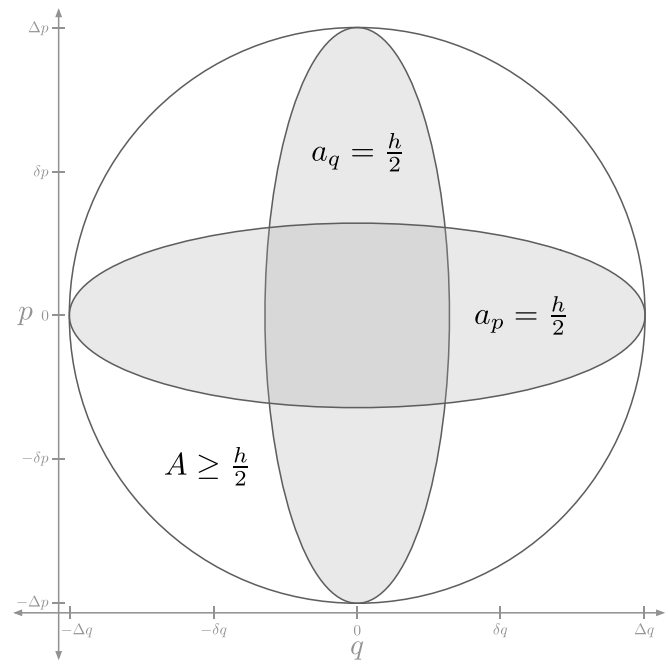

FIG. 1. **The symplectic quantum of action** (schematic, 1D case). The Heisenberg limit $\sigma_q \sigma_p \geq \hbar/2$ maps onto classical semi-axes $\Delta q = \sqrt{2}\sigma_q$, $\Delta p = \sqrt{2}\sigma_p$, projecting the 1σ contour onto a Fermi blob with phase-space area $A = \pi \Delta q \Delta p \geq h/2$. Its interior contains a conjugate pair of quantum blobs, $a_q = \pi\, \delta q\, \Delta p = h/2$ and $a_p = \pi\, \Delta q\, \delta p = h/2$, summing to the quantum of action $h = a_q + a_p$. Their widths are Zurek's reciprocal scales $\delta q = \hbar/\Delta p$ and $\delta p = \hbar/\Delta q$; a_q is squeezed along the position axis and a_p along the momentum axis, and their joint structure is accessible only through ensemble tomography on separate copies of the state. As A grows, a_q and a_p compress along their conjugate axes, enforcing the correspondence principle.

WIGNER NEGATIVITY AND SEMICLASSICAL CAUSTICS

The Wigner–Weyl distribution furnishes an exact phase-space representation of a quantum state ψ,

$$W(q,p) = \frac{1}{\pi\hbar} \int_{-\infty}^{\infty} \psi^*(q+x)\psi(q-x)\, e^{2ipx/\hbar}\, dx, \quad (1)$$

but allows negative values wherever quantum interference is strong [1]. Negativity is not a numerical artifact; it is a structural feature of the Wigner function for any

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pure state beyond the ground state, and no phase-space rotation or scaling removes it.

The semiclassical limit approaches the problem from the opposite direction. The WKB probability density,

$$P_{\mathrm{WKB}}(q) \propto \frac{1}{\sqrt{E - V(q)}}, \tag{2}$$

diverges at the classical turning points $q_\pm$ where $E = V(q)$ and the local momentum vanishes [2, 3]. Physically, the particle slows to rest at the boundary; statistically, it spends infinite time there. The divergence is not regularized by any finite order of the WKB expansion.

Both pathologies share a kinematic root. Wigner negativity arises from interference on scales below $\hbar/\Delta p$, and the WKB caustic arises from assuming $\delta q \to 0$ at the turning point — which demands $\Delta p \to \infty$ and hence unbounded classical action.

THE SYMPLECTIC QUANTUM OF ACTION

De Gosson maps the Heisenberg limit $\sigma_q \sigma_p \geq \hbar/2$ into classical semi-axes $\Delta q = \sqrt{2}\sigma_q$, $\Delta p = \sqrt{2}\sigma_p$, projecting the 1σ contour onto a Fermi blob [7] with phase-space area

$$A = \pi \Delta q \Delta p \geq \frac{h}{2}. \tag{3}$$

Hereafter, Δq and Δp are classical kinematic boundaries.

Zurek's reciprocal scales [5] partition the blob interior: $\delta q = \hbar/\Delta p$ and $\delta p = \hbar/\Delta q$. These define a conjugate pair of orthogonal quantum blobs [6],

$$a_q = \pi\, \delta q\, \Delta p = \frac{h}{2}, \qquad a_p = \pi\, \Delta q\, \delta p = \frac{h}{2}, \tag{4}$$

each carrying the irreducible area $h/2$. The blobs a_q and a_p are orthogonal in phase space — a_q squeezed along the position axis and a_p squeezed along the momentum axis [10] — and their joint structure is accessible only through ensemble tomography on separate copies of the state, never through simultaneous measurement of a single copy. Holding their areas fixed while Δq and Δp expand forces reciprocal compression of δq and δp — a topological necessity enforced by Gromov's non-squeezing theorem [8]. Conjugate pairing locks their areas, $a_q + a_p = h$, to Planck's quantum of action (Fig. 1).

RESOLUTION VIA FINITE CLASSICAL ACTION

The classical action $A = \pi \Delta q \Delta p$ sets the phase-space resolution via the reciprocal scales $\delta q = \hbar/\Delta p$ and $\delta p = \hbar/\Delta q$. The minimum-uncertainty blobs a_q and a_p of area $h/2$ take the Gaussian form of the conjugate squeezed

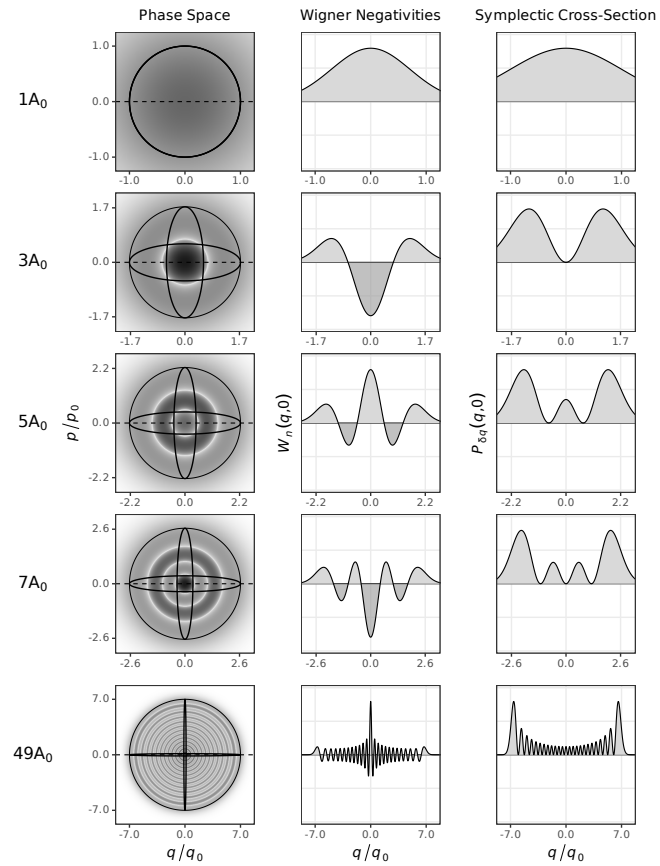

FIG. 2. **Wigner negativity resolved.** Rows show harmonic oscillator eigenstates with action $A = 1, 3, 5, 7, 49\, A_0$ (quantum numbers $n = 0, 1, 2, 3, 24$). All positions in units of q_0, momenta in units of p_0. *Left:* Phase-space Wigner distribution $W_n(q, p)$ with the symplectic quantum of action overlaid. The dashed line marks the $p = 0$ cross-section shown in the adjacent columns. *Center:* Cross-section $W_n(q, 0)$ of the exact Wigner distribution at $p = 0$; dark shading marks negative values, which grow in extent and depth with increasing n. *Right:* Cross-section $P_{\delta q}(q, 0)$ of the convolved phase-space density at $p = 0$; negative regions are eliminated everywhere in phase space by Hudson's theorem, with kernel width set by the finite classical action A. Each row is independently scaled to show shape.

kernels [6, 7],

$$G_{\delta q}(q, p) = \frac{1}{\pi} \exp\left(-\frac{q^2}{\delta q^2} - \frac{p^2}{\Delta p^2} \right), \tag{5}$$

$$G_{\delta p}(q, p) = \frac{1}{\pi} \exp\left(-\frac{q^2}{\Delta q^2} - \frac{p^2}{\delta p^2} \right), \tag{6}$$

where $G_{\delta q}$ is squeezed in δq and spans Δp, and $G_{\delta p}$ is squeezed in δp and spans Δq. Each kernel carries symplectic area $h/2$, saturating the Heisenberg limit. The symplectic positivity condition $\delta q \cdot \Delta p = \hbar$ is satisfied by construction from Zurek's reciprocal scales.

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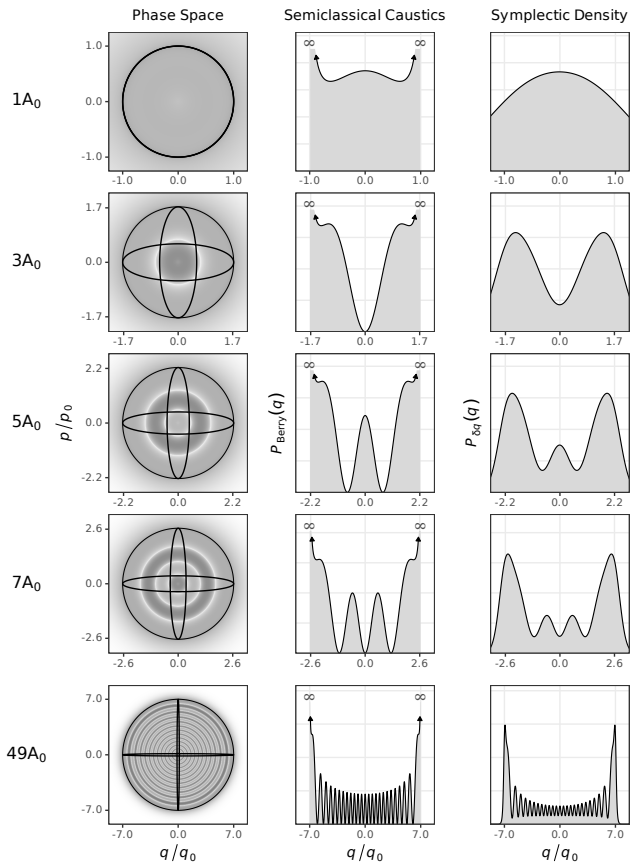

FIG. 3. **Semiclassical caustics resolved.** Rows show the same harmonic oscillator eigenstates as Fig. 2, using Berry's semiclassical phase-space approximation. *Left:* Berry's 2D semiclassical phase-space distribution $W_{\mathrm{Berry}}(q,p)$ with the symplectic quantum of action overlaid. *Center:* Cross-section $W_{\mathrm{Berry}}(q,0)$ at $p=0$; the distribution diverges at the classical turning points $q=\pm\Delta q$ (∞, arrows), a direct manifestation of the semiclassical caustic present at every action level. *Right:* Cross-section $P_{\delta q}(q,0)$ of the convolved phase-space density at $p=0$; enforcing the finite classical action A via the squeezed kernel $G_{\delta q}$ replaces the divergences with finite-amplitude densities, while preserving the quantum oscillation structure in the interior.

The convolved phase-space densities are

$$P_{\delta q}(q,p) = (W * G_{\delta q})(q,p), \qquad P_{\delta p}(q,p) = (W * G_{\delta p})(q,p). \tag{7}$$

Since each kernel satisfies the symplectic positivity condition $\lambda_{\min}(\Sigma + \frac{i}{2}J) = 0$, Hudson's theorem [9] guarantees

$$P_{\delta q}(q,p) \geq 0 \quad \text{and} \quad P_{\delta p}(q,p) \geq 0 \tag{8}$$

pointwise everywhere in phase space. The finite momentum support of $G_{\delta q}$, with width $\Delta p = \hbar/\delta q$, simultaneously bounds the kinematic action at the classical turning points, replacing WKB caustics with finite-amplitude densities.

The squeezed densities differ fundamentally from the Husimi Q-function [4]. Husimi convolves W with a fixed coherent-state kernel set once by the ground state, carrying no information about the system's classical action. The kernels $G_{\delta q}$ and $G_{\delta p}$ are governed entirely by A: as A grows they sharpen anisotropically along the classical trajectory, recovering phase-space structure the fixed Husimi kernel cannot access.

THE HARMONIC OSCILLATOR

The quantum harmonic oscillator provides the canonical test of both resolutions. Its Wigner function is known exactly [1],

$$W_n(q,p) = \frac{(-1)^n}{\pi}\, e^{-(q^2+p^2)}\, L_n\big(2(q^2+p^2)\big)\,, \tag{9}$$

where L_n is the Laguerre polynomial of order n. The Robertson–Schrödinger covariance for eigenstate $|n\rangle$ gives $\sigma_{qq} = \sigma_{pp} = (2n+1)/2$ and $\sigma_{qp} = 0$, yielding classical boundaries $\Delta q = \Delta p = \sqrt{2n+1}$ and reciprocal scales $\delta q = \delta p = 1/\sqrt{2n+1}$, so that $A_{\mathrm{ho}}/A_0 = 2n+1$, where $A_0 = h/2$ is the ground-state action and A_{ho} is the harmonic oscillator classical action. For the harmonic oscillator $A = A_{\mathrm{ho}}$ exactly; the Fermi blob is the energy shell ellipse. All quantities below are expressed in units of the ground-state scales q_0 and p_0, so that $\hbar = q_0 p_0 = 1$.

Wigner Negativity

Figure 2 shows the $p=0$ cross-section $W_n(q,0)$ of the exact Wigner function for five action levels. For $n=0$ the cross-section is a positive Gaussian — the ground state Wigner function is non-negative everywhere in phase space. For $n \geq 1$ the cross-section oscillates and goes negative, with $W_n(0,0) = (-1)^n/\pi$ alternating sign at the origin. The dark-shaded negative regions grow in extent with increasing n, reflecting deepening quantum interference.

The $p=0$ cross-section $P_{\delta q}(q,0)$ of the convolved phase-space density shown in the right column is everywhere non-negative, consistent with Hudson's theorem guaranteeing $P_{\delta q}(q,p) \geq 0$ pointwise throughout phase space. The kernel width $\delta q = 1/\sqrt{2n+1}$ shrinks with increasing action, so at $49A_0$ the convolution barely smooths the oscillations while still eliminating all negativity. Each row is independently scaled; the absolute amplitude of the cross-section decreases with n as the distribution spreads over the growing classical region.

Semiclassical Caustics

Figure 3 shows the same action levels using Berry's semiclassical phase-space approximation [2]. The $p=0$

cross-section $W_{\mathrm{Berry}}(q,0)$ diverges at both classical turning points $q = \pm\Delta q$ for every eigenstate — the caustic is present at all action levels and is not regularized by increasing n. At $49A_0$ the interior oscillations are dense but the turning-point divergences remain.

The $p = 0$ cross-section $P_{\delta q}(q,0)$ of the convolved phase-space density in the right column replaces each divergence with a finite-amplitude peak. The kernel width $\delta q = 1/\sqrt{2n+1}$ provides exactly the resolution needed: coarse at low action where quantum structure dominates, fine at high action where the classical envelope is being recovered. The caustics are resolved without introducing any free parameters — the kernel is fully determined by the system's classical action A_{ho}.

MORSE POTENTIAL

DOUBLE-WELL POTENTIAL

DISCUSSION

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